MIL-G-81065A(OS)
27 March 1967
Superseding
MIL-G-81065(Wep)
21 May 1964

MILITARY SPECIFICATION

GYRO-PENDULUM ASSEMBLY

(For Torpedo Mk 46 Mod 0 and Mk 46 Mod 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of Gyro-Pendulum Assembly, Dwg 2089913, for use in a torpedo. The gyro-pendulum assembly is of one type and class and is r eferred to herein as the gyro-pendulum. (See 6.1 for description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment; General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of

Federal

PPP-B-585	Boxes, Wood, Wirebound

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards; Order of Precedence for the Selection of
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals
MS33586	Metals, Definitions Of Dissimilar

DRAWINGS

Naval Ordnance Systems Command

2089913	Gyro-Pendulum Assembly

PUBLICATIONS

Naval Ordnance Systems Command

WR-11	Design and Test of Packaging, Packing, Shipping, and Handling Equipment for Weapon Systems Components
QAP 60214	Quality Assurance Provisions for Gyro-Pendulum Assembly

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standard for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. c. 20360.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall:

(a) Be manufactured using the procedures proposed for the production lot.

(b) Consist of three gyro-pendulum assemblies.

(c) Be tested as specified in Section 4 herein for the purpose of determining, prior to starting production, that the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the gyro-pendulum shall be in accordance with the requirements specified herein and with the applicable documents listed in Dwg 2089913.

3.2.1.1 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein, and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided herein. AN, JAN, or MS Standard parts shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the AN, JAI?, or MS Standard parts without alteration, and provided the corresponding AN, JAN, or MS Standard part numbers are referenced on the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

3.2.1.2 Fungus proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.2.1.3 Metals. All metals used in the assembly shall be corrosion resistant or shall be adequately protected to resist corrosion during normal service and storage life of the assembly. Aluminum surfaces shall be anodized in accordance with MIL-A-8625, Type I or II as applicable, dyed black.

3.2.2 Gages. The supplier shall provide whatever gages are necessary and adequate to ensure that the material and parts of the gyro-pendulum will meet dimensions and requirements shown on the applicable documents. When requested by the supplier, the procuring activity will furnish existing pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the satisfactory fulfillment of the contract requirements.

3.2.3 Inspections and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities, and shall provide personnel for performing ail acceptance tests specified herein. The test equipment shall be adequate to permit performance of the required acceptance tests for the gyro-pendulum.

3.2.4 Threads. Unless otherwise specified, all. threads shall be in accordance with National Bureau of Standard Handbook H28.

3.2.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals' shall not be used in contact with each other. Dissimilar metals are defined in Standard MS33586.

3.2.6 Safetying. Unless otherwise specified, all threaded parts shall be safetied to withstand all environmental conditions specified herein.

3.2.7 Interchangeability. All parts or subassemblies having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance.

3.2.8 Soldering. All electrical solder connections external to the gyros and pendulums shall be made in accordance with MIL-STD-440. Where necessary, a heat-sink shall be used to prevent deterioration of a component.

3.2.9 Parts list. Each supplier shall provide a parts list and repair instructions with each gyro-pendulum furnished.

3.3 Performance and product characteristics

3.3.1 Physical characteristics. The gyro-pendulum assembly shall conform to the physical requirements delineated in the applicable documents listed on Dwg 2089913.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

3.3.2 Power

3.3.2.1 Gyro motor excitation. The gyro motor power required by the gyro-pendulum shall be single phase alternating current (ac) and shall have the following characteristics:

(a) The frequency shall be 400 plus or minus 20 cycles per second (cps).

(b) The voltage shall be 73.5 plus or minus 5.7 volts peak-to-peak steady state, and shall be capable of 155 volts peak-to-peak for 5 seconds each time the unit is energized. The power source impedance shall be 50 ohms minimum.

(c) The steady state current shall be 475 milliamperes (ma) maximum at a power factor of 0.9 minimum.

(d) Total voltage waveform distortion of 20 percent shall not affect gyro operation.

3.3.2.2 Pickoff excitation. The pickoff (signal circuit) excitation power required by the gyro-pendulum shall be single phase ac having the following characteristics:

(a) The frequency shall be 4800 plus or minus 200 cps.

(b) The voltage shall be 26 plus or minus 2.0 volts root mean square (vrms), and shall be capable of 55 vrms for 5 seconds each time unit energized.

(c) The steady state current shall be 370 ma maximum.

(d) The total harmonic content shall be not greater than 3 percent.

(e) The power factor correction capacitor (C10 on Drawing 2089913) shall be selected so that with 26 plus or minus 0.1 vrms (4.8 kilocycles (kc) plus or minus 10 cps) the current is 290 plus or minus 30 ma at a power factor of 0.8 to 0.96 leading.

3.3.2.3 Signal circuit loading. The signal output external load impedance for each of the five instruments contained in the gyro-pendulum shall be 10,000 plus or minus 500 ohms resistive, shunted by 900 to 1100 micromicrofarads (μμfd).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

3.3.2.4 Insulation resistance. The insulation resistance between isolated circuits and between circuits and mounting surface shall be not less than 50 megohms at 500 volts direct current (de) for a period of 15 seconds. The above voltage shall not be applied more than three times to any circuit.

3.3.3 Performance characteristics

3.3.3.1 Power and loading. Whenever electrical operation is indicated in the following paragraphs of Section 3, excluding Table 1, conditions specified in 3.3.2 shall be provided.

3.3.3.2 Phase relationship of signal voltages. The phasing of the signal voltages shall be as designated on Drawing 2089913.

3.3.3.3 Signal phase shift. The phase shift between pickoff excitation voltage and signal (output voltage of each of the five instruments contained in the gyro-pendulum shall not exceed zero plus or minus 5 degrees or 180 plus or minus 5 degrees for output voltages of 1.00 to 15.00 vrms.

3.3.3.4 Static null of signal voltages. With the gyro-pendulum motionless and oriented with respect to gravity as shown on Drawing 2089913 within 0.1 degree, and with the, gyro motors running, each of the five signal output voltages shall not exceed 125 millivolts rms.

3.3.3.5 Distortion. The total distortion of each of the five outputs shall not exceed five percent for output voltages of one vrms or greater or 10 percent for output voltages of 0.5 to 0.9 vrms. The above is referenced to the excitation voltage at a source impedance of 50 ohms.

3.3.3.6 Natural frequency. The undamped natural frequency of the gyro or pendulums shall be 27 plus or minus 5 cps.

3.3.3.7 Damping ratio. The damping ratio shall be 0.68 plus or minus 0.15 at 75 degrees Fahrenheit (F) and 0.35 to 2.0 from O degrees F to 160 degrees F.

3.3.3.8 Linear acceleration sensitivity. Errors due to linear acceleration along any gyro axis shall not exceed 0.2 degree/see/g.

3.3.3.9 Rate gyro spin-up time. The time required for each of the three gyros to attain synchronous speed after application of motor power shall not exceed seven seconds.

3.3.3.10 Outputs. The rate gyro outputs shall be not greater than directly proportional to the 400 cps frequency but shall not be affected by the specified 400 cps voltage variation. The pickoff outputs of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

five instruments shall not vary more than plus or minus 3 percent, due to the allowable frequency variation of the 4800 cps, but shall be directly proportional to the 4800 cps excitation voltage amplitude.

3.3.3.11 Instrument performance characteristics. Table I further defines rate gyro and pendulum specifications as functions of forces applied to the gyro-pendulum in the appropriate axes. Unless otherwise specified, the following conditions apply:

(a) 400 plus or minus 2 cps at 26 plus or minus 0.1 vrms

(b) 4800 plus or minus 10 cps at 26 plus or minus 0.1 vrms

(c) 10 plus or minus 0.1 kilohms resistive load, shunted by 900 to 1100 μμfd.

(d) Temperature 0 degrees F to plus 160 degrees F (the requirements of Table I are to be met over the entire temperature range)

3.3.4 Life. Life requirements shall be as specified for the following unit conditions (see 6.3).

3.3.4.1 Service life. The gyro-pendulum shall meet the requirements of 3.3.1 through 3.3.3.10 throughout its mission time of nine minutes.

3.3.4.2 Total operating life. The gyro-pendulum shall be capable of satisfactory operation for a minimum 500 total hours of gyro motor operation from date of acceptance by the procuring activity.

3.4 Environmental requirements. Unless otherwise specified, the gyro-pendulum shall be capable of conforming to the performance and product characteristics requirements of this specification after being exposed, at any temperature in the applicable range, to the environments under all unit conditions specified in this section.

3.4.1 Vibration

3.4.1.1 The gyro-pendulum shall be capable of withstanding the vibrations shown in Table II.

3.4.1.2 Vibration, packaged and packed. The gyro-pendulum, after being subjected to the tests specified in 4.5.2.1.2, shall show no damage and shall meet all the requirements of Section 3.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:58Z
Check the source to verify that this is the current version before use.

Table I

Rate Gyros		Characteristic	Pendulums	
Quantity or Limit	Condition		Quantity or Limit	Condition
14.2 ±0.4 vrms	40 ±0.2 degrees per sec. angular velocity	Signal output voltage	14.2 ±0.4 vrms	One ±0.008g
±0.3 vrms max. allowable change in signal output voltage	Resistive portion of load changed by ±500 ohms at 40 ±0.2° per sec. Capacitive portion of load changed by ±100 mmfds.	Output Regulation	±0.3 vrms max. allowable change in signal output	Resistive portion of load changed by ±500 ohms at 1 ±0.008g. Capacitive portion of load changed by ±100 mmfds.
±75 mvrms max.	1 to 20° per sec.	Linearity (Independent base)	±75 mvrms	0.0175 g to 0.5 g
±150 mvrms max.	20 to 30° per sec.		±200 mvrms max.	0.5 to 1 g
±300 mvrms max.	30° per sec. to stop		±350 mvrms max.	1 g to stop
0.02° per sec. max.		Resolution	0.002 g max.	
0.03° per sec. max.		Sensitivity Threshold	0.003 g max.	
0.12° per sec.	0 to +40 to 0 to -40 to 0 ±0.2 degree per sec.	Hysteresis	0.003 g	0 to +1 to 0 to -1 to 0 to ±0.008 g

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

Table I (Continued)

Rate Gyros		Characteristic	Pendulums	
Quantity or Limit	Condition		Quantity or Limit	Condition
At 54 $^{+10}_{-0}$ deg/sec.		Mechanical stops	At 2.25 $^{+0.4}_{-0}$ g	
800° per sec.	No change in characteristics	Applied force (maximum 5 minutes)	35 g	No change in characteristics
100 mvrms max.	Static	Null (Including Hysteresis)	100 mvrms max signal output voltage	Static
0 ±4 degrees or 180 ±4 degrees	3 to 40 ±0.2° per sec.	Phase shift between pick-off excitation voltage and signal output voltage	0 ±4 degrees or 180 ±4 degrees	0.05 to 1 ±0.008 g

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

Table II

Vibration Schedule

Unit Condition	Double Amplitude in. minimum	Acceleration (g) minimum	Frequency cps
Energized	0.75		5-6
		1.3	6-16
	0.1		16-24
		3.0	24-40
	0.036		40-74
		10.0	74-500

3.4.2 Shock

3.4.2.1 The gyro-pendulum shall be capable of withstanding shock as specified in Table III.

Table III

Shock Schedule

Unit Condition	Peak Magnitude (g's)	Minimum Duration (ms)	Acceleration Direction
Nonoperating Ready, non-operating	60	8.0	Any
Ready	45 85 300	20.0 0.5 0.5	Fwd Any Aft

3.4.2.2 Shock; transportation, storage, and handling. The gyro-pendulum, after being subjected to the test specified in 4.5.2.2.2, shall show no damage and shall meet all the requirements of Section 3.

3.4.3 Acceleration. The gyro-pendulum shall be capable of withstanding the accelerations specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

Table IV

Acceleration Schedule

Condition	Magnitude (g)	Acceleration Direction
Nonoperating	25	Forward
Ready, non-operating Ready	35	Aft

3.4.4 Temperature. The gyro-pendulum shall be capable of conforming to the requirements of this specification at any temperature within the limits of Table V, in the unit conditions specified in Table V.

Table V

Temperature Schedule

Condition	Temperature Range
Nonoperating	-45 to +140 degrees F
Ready, nonoperating	0 to +140 degrees F
Ready and operating	0 to +160 degrees F

3.4.5 Pressure. The gyro-pendulum shall be capable of withstanding a pressure of 2.7 psia.

3.4.6 Humidity. The gyro-pendulum shall withstand exposure to 95 percent relative humidity at 110 degrees F for eight hours without degradation of the finish or functional capabilities.

3.4.7 Storage orientation sensitivity. The rms static null signal output of each of the five instruments (see 6.1.2) shall not exceed 100 millivolts (rev) after the gyro-pendulum has been stored in any 90 degree pitch or roll orientation for one year. Conformance shall be demonstrated by a 48 plus or minus 1 hour storage test made at any time during the one year storage period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

3.6 Identification. The gyro-pendulum shall be permanently identified in accordance with MIL-STD-130 and the applicable documents listed in Dwg 2089913.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and service conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the gyro-pendulum shall be classified as follows:

(a) Preproduction inspection

(1) Reproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Acceptance inspection

(l) Acceptance sampling (see 4.3.3)

(2) Acceptance tests and examinations (see 4.7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

4.3 Sampling

4.3.l preproduction sample. A preproduction sample of three gyro-pendulums manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of' the preproduction sample shall be based on no defects in the sample. Further production of the gyro-pendulum by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production tests detailed in 4.6 shall be performed at an activity designated by the procuring activity. Samples shall consist of units that have satisfactorily passed the acceptance tests and examinations detailed in 4.7, selected as follows:

(a) Step 1. Upon initiation of production, randomly select 5 units from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to Step 2.

(b) Step 2. Randomly select 5 units from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to Step 3.

(c) Step 3. Randomly select 5 units from each group of 50 units. Continue with Step 3 for the balance of the contract or until a defective unit is found.

4.3.2.1 Defective units. When a defective unit is found in any group, the supplier shall:

(a) Determine the cause of the defect.

(b) Determine if the defect is an isolated case.

(c) Take necessary action to eliminate the defects from units on hand and, if so directed by the procuring activity, from those units previously shipped.

(d) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(e) Return to Step 1 (4.3.2(a)) and proceed as indicated for inspection of units subsequently produced.

Source: https://assist.dla.mil -- Downloaded: 202640-23T04:58Z
Check the source to verify that this is the current version before use.

4.3.3 Acceptance sampling. The tests and examinations specified in 4.7 shall be conducted on each gyro-pendulum to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the gyro-pendulum shall be subjected to tests under the following conditions:

(a) Temperature	Room ambient (60 to 95 degrees F)
(b) Humidity (relative)	Room ambient, up to 95 percent
(c) Altitude	Normal ground atmospheric pressure

4.5 Preproduction test. The gyro-pendulum with the preproduction sample (identified as gyro-pendulums 1, 2 and 3) shall be subjected to the tests specified in Table VI. Certified inspection data covering the results of preproduction tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Date of test and names of persons conducting the tests.

(b) Run or test plan used.

(c) Serial number of gyro-pendulum.

(d) Test results of preproduction tests (4.5.2 through 4.5.3).

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein:

4.5.1.1 A vibration table capable of vibrating a 5 pound load, plus fixture weight, at 10 g’s maximum over a frequency range of 5 to 500 cycles per second with a maximum double amplitude of 0.75 inch.

4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g’s maximum at durations from 0.5 to 21 milliseconds.

4.5.1.3 A centrifuge capable of exerting acceleration forces up to 35 g’s magnitude for periods of three minutes minimum.

4.5.1.4 A temperature chamber capable of programmed temperature variations over a range of minus 50 to plus 145 degrees F.

4.5.1.5 A humidity chamber capable of providing a constant humidity condition of 95 percent at 110 degrees F for extended periods up to 8 hours .

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

4.5.1.6 Items of test equipment specified in QAP 60214.

Table VI

Preproduction Test Schedule

Test Sequence	Test Description	Gyro-Pendulum 1	2	3
1	Acceptance tests and examinations (4.7)	x	x	x
2	Vibration (4.5.2.1)			x
3	Shock (4.5.2.2)	x		
4	Acceleration (4.5.2.3)			x
5	Temperature, storage (4.5.2.4.1)		x	
6	Temperature, operating (4.5.2.4.2)	x		
7	Pressure test (4.5.2.5)		x	
8	Humidity test (4.5.2.6)			x
9	Life test (4.5.2.7)	x		
10	Storage orientation sensitivity test (4.5.2.8)	x	x	x
11	Acceptance tests and examinations (4.7)	x	x	x

4.5.2 Detailed preproduction tests

4.5.2.1 Vibration

4. 5.2.1.1 The gyro-pendulum shall be vibrated for a total of nine hours (3.4.1, and Tables II and VII) in accordance with the requirements of Procedure XII of MIL-E-5272 and the vibration deviations of 4.5.2.1.1.1.

4.5.2.1.1.1 Vibration deviations. The following are deviations from Procedure XII of MIL-E-5272. Individual resonance surveys shall be conducted with vibration applied along each of three mutually perpendicular axes. The resonant and cycling periods in each axis shall be three hours at room temperature in accordance with Table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

Table VII

Resonant and Cycling Schedule

No. of Resonances	Time at Resonance*	Cycling Time
0	0	180
1	27	153
2	54	126
3	81	99
4	108	72

*Twenty-seven minutes at each resonance.

4.5.2.1.2 Vibration; packaged and packed. The gyro-pendulum; packaged and packed as specified in Section 5, shall be vibration tested as specified in Weapons Requirement WR-11.

4.5.2.2 Shock

4.5.2.2.1 Mount the gyro-penculum onto a shock test fixture so that minimum of 90 percent of the table input of the fixture will be applied to the gyro-pendulum under test. Subject the gyro-pendulum to the shocks listed in Table VIII (in the condition specified in Table III) as specified in the Shock Test (3.4.2). Duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock. Test equipment is not restricted to that specified in MIL-E-5272.

Table VIII

Shock Schedule

Peak Magnitude (g's)	Duration (ms)	Impact Direction	Number of Pulses
300	0.5 to 1.0	Longitudinally Aft	3
85	0.5 to 1.0	Any direction except Longitudinally Aft	6
60	8.0 to 8.5	Any	9
45	20.0 to 21.0	Longitudinally Fwd	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

4.5.2.2.2 Shock; transportation, storage and handling. The gyro-pendulum, packaged and packed as specified in Section 5, shall be subjected to the appropriate shock isolation tests specified in Weapons Requirements WR-11.

4.5.2.3 Acceleration. The gyro-pendulum shall be subjected to the acceleration magnitudes and directions required by Table IV for three minutes at room temperature for each condition specified.

4.5.2.4 Temperature

4.5.2.4.1 Temperature, storage. The gyro-pendulum, preserved and packaged in accordance with 5.2.1, shall be placed in a temperature controlled chamber and subjected to the following temperature cycles:

(a) Heat to plus 140 plus 5 minus 0 degrees F in 2.5 plus or minus 0.5 hours.

(b) Maintain at plus 140 plus 5 minus 0 degrees F for 4 hours minimum.

(c) Cool to minus 45 plus 5 minus 5 degrees F in 2.5 plus or minus 0.5 hours.

(d) Maintain at minus 45 plus O minus 5 degrees F for 4 hours minimum.

(e) Return to room temperature in 2.5 plus or minus 0.5 hours.

Repeat this schedule for a minimum of three cycles.

4.5.2.4.2 Temperature, operating. The gyro-pendulum shall be stabilized at a temperature of zero degrees F and tested to conform to the requirements of 3.3.3.11. The gyro-pendulum shall then be stabilized at plus 160 degrees F and again tested to conform to the requirements of 3.3.3.11.

4.5.2.5 Pressure test. The gyro-pendulum shall be subjected to pressure of 2.7 psia.

4.5.2.6 Humidity test. The gyro-pendulum shall be placed in a suitable test chamber and be subjected to 95 percent relative humidity for a period of eight hours minimum at a temperature of 110 degrees F minimum.

4.5.2.7 Life test. The gyro-pendulum shall be operated for a total of 500 hours. The maximum duty cycle shall be three hours on and one hour off under such ambient conditions that the gyro-pendulum will cool to room temperature within the one hour off time.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:58Z
Check the source to verify that this is the current version before use.

4.5.2.8 Storage orientation sensitivity test. Store the gyro-pendulum for 48 plus or minus 1 hours in each of the following orientations: Pitch down, pitch up, cw roll, ccw roll. Immediately after each 48 plus or minus 1 hour storage orientation measure the rms static null signal output of each of the five instruments in accordance with QAP 60214, paragraph 20.3.2.

4.5.3 Preproduction acceptance criteria. Acceptance shall be based on satisfactory completion of test sequence items 1 through 11 of Table VI.

4.5.4 Functional tests. At the conclusion of the tests specified in 4.5.2.1.2 and 4.5.2.2.2 above, functional and other tests, as applicable, shall be conducted on the gyro-pendulum to determine freedom from operational malfunction and compliance with all requirements of Section 3.

4.5.5 Waiver of tests. Tests specified in 4. 5.2.1.2 and 4.5.2.2.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.6 Periodic Production tests. The periodic production sample, selected in accordance with 4.3.2 shall be subjected to the tests of 4.6.2.

4.6.1 Periodic Production test equipment. The following items of test equipment are required to perform the periodic production test specified herein.

(a) Items of test equipment specified in QAP 60214.

(b) Complex wave vibration machine (Rotocon Model RC-3 50/2000 or equivalent). The vibration machine shall be capable of vibrating a weight of 10 pounds at a fundamental frequency of 50 cycles per second with a peak magnitude of 4.5 plus or minus 0.5 g. White noise acceleration density (GO) shall be 0.006 g^2/cps, and bandwidth shall be 2000 cps.

4.6.2 Periodic Preduction test schedule. Subject the gyro-pendulum to the following tests in the order indicated:

(a) Vibrate the gyro-pendulum, nonoperating and in one (optional) axis only, for a period of 4 to 5 minutes on the complex wave vibration machine specified in 4.6.1(b).

(b) Acceptance tests and examinations (see 4.7).

4.7 Acceptance tests and examinations. Unless otherwise specified each gyro-pendulum delivered under contract shall be subjected to the tests and examinations contained in QAP 60214. Failure to meet any requirement specified herein shall be considered cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:58Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.1 Cleaning, drying, and preservative application. Not applicable.

5.1.1.2 Unit packaging. Unit packaging of the gyro-pendulum shall be in accordance with Method IIb of MIL-P-116. Each unit package shall contain one gyro-pendulum.

5.1.1.3 Intermediate package. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning, drying, and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging of the gyro-pendulum shall be in accordance with Method III of MIL-P-116. Each unit package shall contain one gyro-pendulum.

5.1.3.3 Intermediate package. Not applicable.

5.2 Packing

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with MIL-E-17555.

5.2.1.1 Exterior containers. The gyro-pendulum, packaged as described in 5.1.1.2, shall be packed in a wood box in accordance with PPP-B-621, Class 2, or PPP-B-601, overseas type, or in a wirebound box in accordance with PPP-B-585, Class 3 (weight limit not to exceed 150 lbs) or a fiberboard box in accordance with PPP-B-636, Class 2, or PPP-B-591, overseas type, (weight limit not to exceed 70 lbs).

5.2.2 Level B. In addition to the following, Level B packing shall be in accordance with MIL-E-17555.

5.2.2.1 Exterior containers. The gyro-pendulum packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with PPP-B-621, Class 1, or PPP-B-601, domestic type, or in a wirebound box in accordance with PPP-B-585, Class 1, (weight limit not to exceed 150 lbs), or in a fiberboard box in accordance with PPP-B-636, Class 1, or PPP-B-591, domestic type (weight limit not to exceed 70 lbs).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The gyro-pendulum, packaged in accordance with 5.1.3.2; shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Levels A, B, or C cushioning. The unit package or pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure the gyro-penculum will meet all the requirements of Section 3 after being tested in accordance with 4. 5.2.1.2 (Vibration, packaged and packed) and 4.5.2.2.2 (Shock, transportation, storage, and handling). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use and description

6.11 Intended use. The gyro-pendulum is intended for use in a torpedo control system. The unit provides Yaw Rate, Pitch Rate, and Roll Rate stabilization voltages and also provides Pitch and Roll angular position voltages.

6.1.2 Description. The gyro-pendulum is an electro-mechanical assembly mounting three rate gyros and two restrained pendulums wired to an electrical connector. The physical configuration conforms to the applicable documents listed on Dwg 2089913.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Assigned activity for preproduction and periodic tests.

(e) Data required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

(f) Any exceptions to this specification.

(g) All additional samples required shall be provided and inspected at the expense of the supplier, when such samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116 and to 4.5.5.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition

6.3.1.1 Nonoperating. The condition Of the uninstalled gyro-pendulum when stored, transported, or handled in a packaged condition.

6.3.1.2 Ready, nonoperating. The condition of the gyro-pendulum when assembled with the end item, transported, and handled in an unpackaged condition.

6.3.1.3 Ready. The condition of the gyro-pendulum when fully assembled with the end item and installed on the launching mechanism prior to operating.

6.3.1.4 Operating. Operating is the condition of the gyro-pendulum when performing its required function.

6.3.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or “to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g’s). The frequency is given in cycles per second (cps).

6.3.3 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g’s), duration in units of time (milliseconds (ins)), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

6.3.4 Acceleration (Steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g’s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

6.3.5 Temperature. The temperature of the environment to which the gyro-pendulum may be exposed, expressed in degrees Fahrenheit (°F).

6.3.6 Pressure. The pressure of the environment to which the gyro-pendulum may be exposed, expressed in pounds per square inch absolute (psia).

6.3.7 Humidity. The relative humidity of the environment to which the gyro-pendulum is exposed, expressed as a percent.

6.3.8 Linearity (independent based). The maximum allowable deviation of the output voltage versus input acceleration or rate curve from the best straight reference line.

6.3.9 Resolution. The smallest change in input acceleration or rate that is required to produce an incremental change in output voltage at any point throughout the entire range of input acceleration or rates other than from a zero input acceleration or rate.

6.3.10 Sensitivity threshold. The smallest input acceleration or rate that is required to produce an initial change in output voltage (starting from zero acceleration or rate).

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

6.4.1 Level A. This level shall be used for those items which are to be shipped to intermediate destinations or stored under indeterminate conditions for redistribution anywhere.

6.4.2 Level B. Not applicable.

6.4.3 Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-G-81065A(OS) GYRO-PENDULUM ASSEMBLY (For Torpedo Mk 46 Mod 0 and Mk 46 Mod 1)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8370

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL ORDNANCE SYSTEMS COMMAND (ORD 9343)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:58Z
Check the source to verify that this is the current version before use.